

Natanael Albert

Yogyakarta, Indonesia

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Profile

Recent Information Technology graduate from Universitas Gadjah Mada, currently based in Kudus and available for full-time employment in Kudus, with strong interests and skills in computer programming and analytical problem-solving. Experienced in data processing and transformation (ETL) using Python, time-series-based machine learning, and application and API development for data integration between systems, developed through an industry internship and team projects. Actively involved in computer and mathematics competitions as both a participant and technical committee member. Responsible, communicative, and accustomed to working effectively in teams.

Technical Skills

Data Processing and Transformation (ETL): Python (Pandas), data cleaning and exploration, data integration across sources

Programming Languages: Python, C#, C, C++, JavaScript, TypeScript

Databases and APIs: Oracle Database, PostgreSQL, SQL, Prisma ORM, Postman

Artificial Intelligence and Data Science: Keras, Deep Learning, LSTM, time-series analysis

Application Development: ASP.NET, PHP, AJAX, Laravel, Next.js, Kotlin

Software Testing: Black Box Testing, Gray Box Testing

Version Control: Git

Education

Universitas Gadjah Mada Aug 2022 – 2026
B.Sc. in Information Technology | GPA: 3.62/4.00 Yogyakarta, Indonesia

SMAN 1 Kudus 2019 – 2022
Mathematics and Natural Sciences Kudus, Indonesia

Experience

Senior Information Technology Project Teaching Assistant Jan 2026 – Jul 2026
Department of Electrical Engineering and Information Technology, Universitas Gadjah Mada Yogyakarta, Indonesia

- Supervised and guided students throughout the learning process and module assignments.
- Evaluated students' module and practical assignment submissions.
- Explained concepts and provided technical problem-solving support during and outside practical sessions.

Web Application Developer Intern Jan 2025 – Feb 2025
PT Pura Barutama Kudus, Indonesia

- Developed an ASP.NET-based web application to digitize the company's reporting system.
- Integrated the application with an Oracle database using custom procedures for structured data retrieval.
- Conducted black box and gray box testing to ensure system functionality, security, and usability.
- Repository: github.com/NatanaelAlbert22/Pura-QPro-Mobile

Staff and Coordinator of Competitive Programming Division Nov 2022 – May 2024
FindIT! Yogyakarta, Indonesia

- Coordinated with problem setters to develop programming problems for competitions.
- Assisted in designing and preparing programming problems for competitions.
- Set up servers and websites for competitive programming competitions.

Projects

Berikut versi Inggrisnya. Saya buat tetap **professional dan ATS-friendly**, sekaligus mempertahankan angka dan istilah teknisnya.

“`latex`

KosBudget Team Project
Financial Management Application for Boarding House Students | Python, Streamlit

- Developed a web-based financial management application that helps users automatically allocate monthly budgets across different expense categories.
- Designed a decision-scoring logic to prioritize expense categories based on urgency, frequency, and impact if unmet.
- Built authentication, expense tracking, and financial visualization dashboard modules using Streamlit.

- Repository: github.com/Drafaund/KosBudget

HarvestX

Team Project

Food Price Analytics and Forecasting Dashboard | Next.js, TypeScript, PostgreSQL, Prisma

- Built Next.js API routes to retrieve historical data, calculate monthly average prices, and integrate them with forecasting results.
- Designed a PostgreSQL data schema using Prisma ORM to store historical data and commodity price forecasts.
- Developed a web dashboard to monitor daily food prices, historical trends, five-day forecasts, and estimated spending simulations based on commodities, regions, and market types.
- Repository: github.com/JosuaAdhiCandraN/product1.1

E-commerce Customer Retention Analysis

Individual Project

SQL Analysis and BI Dashboard for E-commerce Customer Retention | Python, DuckDB, SQL, Tableau

- Analyzed 98,198 e-commerce transactions using advanced SQL (CTEs and window functions) to identify stagnant revenue growth over an eight-month period in 2018.
- Built cohort retention analysis and RFM (Recency, Frequency, Monetary) customer segmentation for nearly 95,000 unique customers to identify retention budget priorities.
- Identified and corrected bias in RFM frequency scores caused by tied values, which had previously resulted in 37% of customers being misclassified as “Loyal Customers.”
- Designed an interactive Tableau Public dashboard presenting revenue trends, retention heatmaps, and customer segment distributions for non-technical stakeholders.
- Repository: github.com/USERNAME/ecommerce-customer-retention-analysis

Feature A/B Test & Funnel Analysis

Individual Project

Statistical Experimentation for Mobile Game Feature Testing | Python, SciPy, Statsmodels

- Designed and executed a complete A/B testing experiment involving 90,189 users, including power analysis, proportion tests (z-test and chi-square), bootstrap confidence intervals, and multiple testing correction (FDR).
- Found a statistically significant 0.83 percentage-point decrease in Day-7 retention for the treatment group ($p=0.0026$ after FDR correction), while Day-1 retention showed no significant difference.
- Validated sufficient sample size through power analysis, achieving 89.2% statistical power, enabling non-significant results to be distinguished from insufficient sample size.
- Developed statistically grounded go/no-go recommendations for the product team, including documentation of methodological limitations such as Sample Ratio Mismatch and limitations of CUPED applicability.
- Repository: github.com/USERNAME/feature-ab-test-funnel-analysis

Thesis

Transfer Learning for Time-Series-Based Air Quality and Climate Prediction with Limited Yogyakarta Data

2025 – 2026

Undergraduate Thesis | Python

- Developed a time-series forecasting system for air quality (PM2.5/PM10) and climate variables (temperature/humidity) using limited Yogyakarta data and a transfer learning approach.
- Applied Long Short-Term Memory (LSTM) with domain adaptation mechanisms (CORAL and Deep CORAL) to bridge differences in data distributions across cities.
- Achieved competitive evaluation results and demonstrated the effectiveness of transfer learning for climate-related forecasting.

Certifications

- DataCamp: ETL and ELT in Python – Nov 2024
- DataCamp: Exploratory Data Analysis in Python – Sep 2024
- DataCamp: Cleaning Data in Python – Sep 2024
- DataCamp: Introduction to Statistics in Python – Aug 2024
- DataCamp: Advanced Deep Learning with Keras – May 2024
- DataCamp: Introduction to Deep Learning with Keras – May 2024

Achievements

- 1st Place, National Science Competition (KSN) in Computer Science, Kudus Regency/City (2020, 2021).
- National-Level Finalist, EMERALD EDUCATION CENTER Mathematics Competition (2018).